🧬 Deep-Regulatory: Deciphering Cell-Specific Transcription Factor Binding & Chromatin Accessibility

📝 Research Overview

The fundamental challenge in genetics is understanding why the same DNA sequence behaves differently across various cell types (e.g., Neurons vs. Cardiomyocytes). This project presents a hybrid **Deep Learning** framework that analyzes DNA sequences and epigenetic signatures to predict regulatory "hotspots." The goal is to decode the functional impact of non-coding mutations (SNPs) within a biological context.

⚠️ The Biological Problem

- **Context-Specific Regulation:** 98% of the human genome is non-coding, where enhancers and promoters control gene expression. Their activity is cell-specific and identifying them experimentally is expensive and time-consuming.
- **Variant Interpretation:** Single Nucleotide Polymorphisms (SNPs) often reside in regulatory regions, but predicting their ripple effect across a gene regulatory network remains a significant challenge.

✅ The Solution: A Hybrid Pipeline

We designed a multi-stage pipeline that bridges computational predictions with experimental validation:

1. In Silico Modeling (Dry-Lab)

- **Transformer Architecture:** Developed a sophisticated model to process DNA sequences (1000bp windows) alongside epigenetic marks such as **H3K27ac** and **DNA Methylation**.
- **XAI Integration:** Leveraged **Integrated Gradients** and **Grad-CAM** to map AI decisions, enabling the identification of specific Transcription Factor (TF) binding motifs.
- **In Silico Mutagenesis:** Automated SNP impact analysis to determine the functional importance of every base pair.

2. In Vitro Simulation (Wet-Lab Bridge)

To verify AI predictions, we engineered virtual validation modules:

- **Luciferase Reporter Assays:** Quantifying the strength of predicted enhancers.
- **CRISPRi / RT-qPCR:** Simulating the silencing of specific regulatory elements to quantify the resulting decrease in target gene expression.

📊 Dataset & Features

- **Data Source:** High-fidelity synthetic genomic data simulating the complexity of **ENCODE** and **NCBI GEO** datasets.
- **Features:**
  - One-hot encoded DNA sequences.
  - Cell-specific epigenetic signal tracks (e.g., Chromatin accessibility).
  - Ground-truth Transcription Factor (TF) binding labels.

🛠️ Tech Stack

- **Deep Learning:** TensorFlow / Keras (Functional API), PyTorch.
- **Genomics Tools:** BioPython, Logomaker (for sequence logos), Scikit-learn.
- **Data Science:** NumPy, Pandas, Matplotlib, Seaborn.
- **Explainable AI:** Custom GradientTape implementations for saliency mapping.

🚀 Key Research Features

- **Multi-Modal Integration:** Combined analysis of DNA sequence and epigenetic signals.
- **High-Throughput VEP:** Variant Effect Prediction, calculating pathogenicity scores for SNPs.
- **Visual Research Dashboard:** Integrated Sequence Logos and virtual Western Blot/Gel simulations ready for clinical presentation.

🔮 Future Research & Impact

- **Precision Medicine:** Creating personalized regulatory maps for cancer genetics and neuro-developmental disorders.
- **Drug Discovery:** Designing ligands that specifically target mutated regulatory regions.
- **Live Validation:** Deploying this pipeline on real-world **ChIP-seq** and **ATAC-seq** data.

📁 Repository Structure

```
├── Genomic_Regulation_Transformer.ipynb   # Main Research Notebook
├── simulations/                   # Virtual Wet-Lab Results (Gel/qPCR)
├── models/                        # Trained Transformer Weights
```

└── README.md # Project Documentation